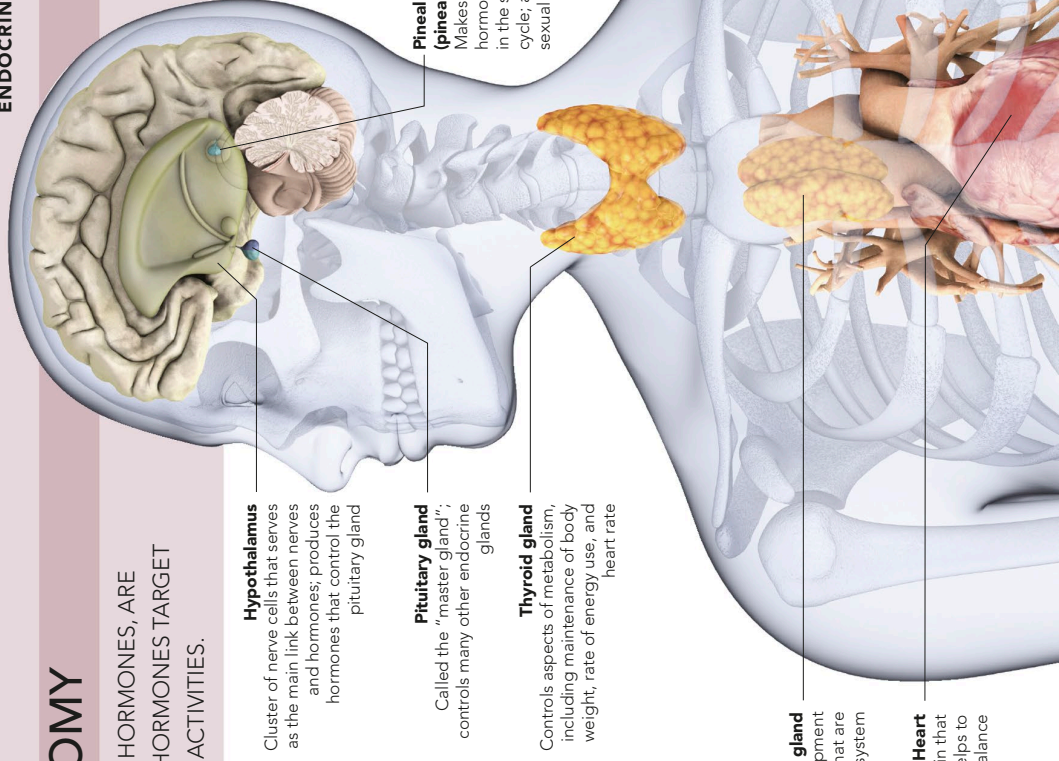


ENDOCRINE ANATOMY

THE BODY'S CHEMICAL MESSENGERS, HORMONES, ARE PRODUCED BY ENDOCRINE GLANDS. HORMONES TARGET CERTAIN TISSUES TO REGULATE THEIR ACTIVITIES.

The endocrine system is composed of bodies of glandular tissue, such as the thyroid, and also includes glands within certain organs, including the testes, ovaries, and heart. These glands and tissues secrete hormones that control and coordinate body functions such as the breakdown of chemical substances in metabolism, fluid balance and urine production, growth and development, and sexual reproduction. As hormones travel in the blood, each hormone reaches every body part. However, the specific molecular shape of each hormone slots only into receptors on its target tissues or organs.



Hypothalamus
Cluster of nerve cells that serves as the main link between nerves and hormones; produces hormones that control the pituitary gland

Pituitary gland
Called the "master gland"; controls many other endocrine glands

Thyroid gland
Controls aspects of metabolism, including maintenance of body weight, rate of energy use, and heart rate

Thymus gland
Produces hormones involved in development of white blood cells, called T cells, that are part of the immune system

Heart
Produces a hormone called atriopeptin that reduces blood volume and pressure and helps to regulate fluid balance

Pineal gland (pineal body)
Makes melatonin, a hormone important in the sleep-wake cycle; also influences sexual development